

Medicare Cardiovascular Spending Variation by Anxiety Status

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Source data. CMS DE-SynPUF Medicare claims.

Population. Beneficiaries with hypertension, coronary artery disease, or atrial fibrillation.

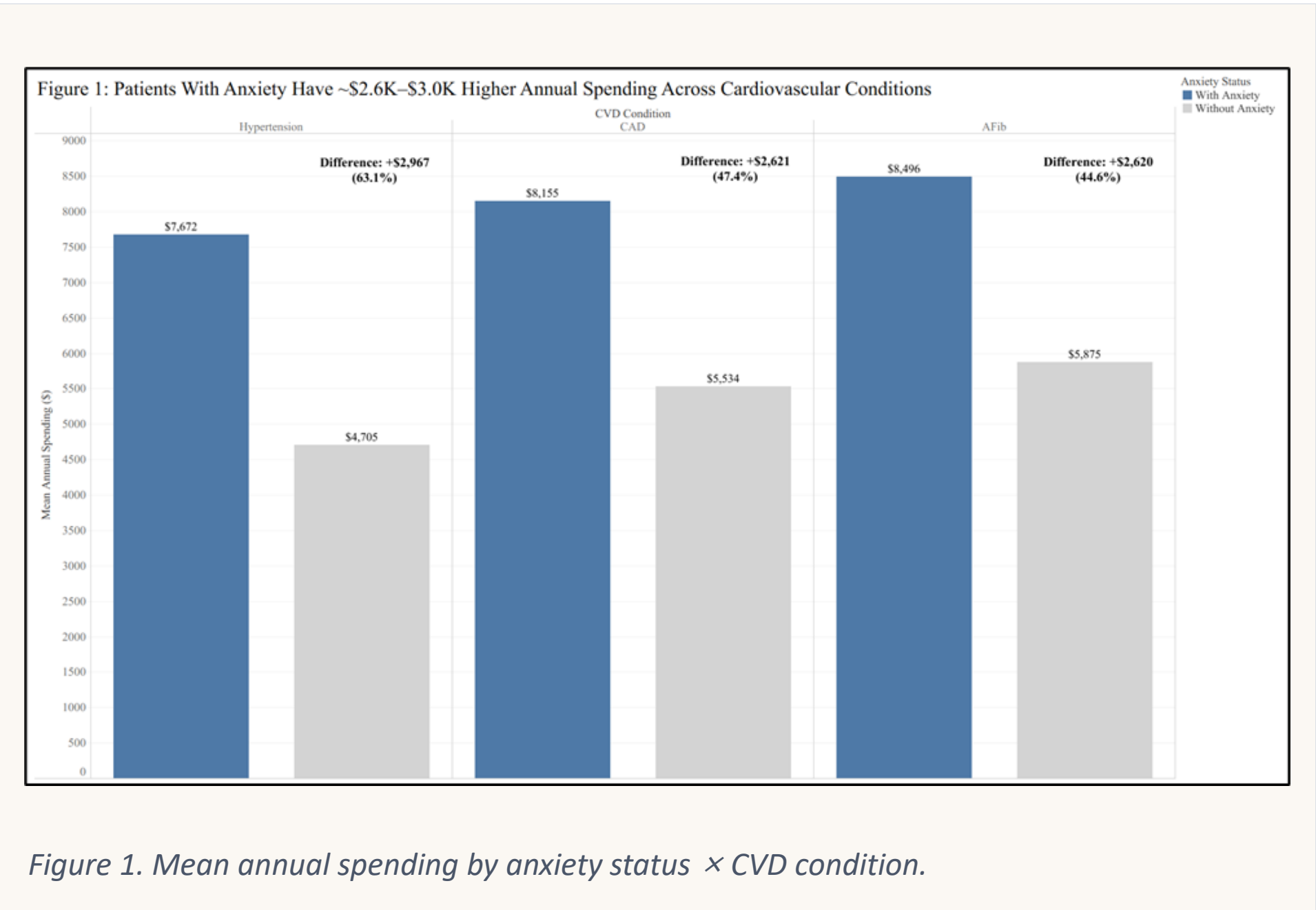
Question. Does anxiety independently raise cardiovascular spending, and where does the extra cost go?

KEY TAKEAWAY

Anxiety is associated with **7.8% to 12.4% higher** Medicare spending across cardiovascular conditions. The gap is **driven primarily by outpatient care** and consistent across all 50 states and DC.

CONDITIONS	METHOD	GEOGRAPHY
Hypertension · CAD · AFib	Adjusted Gamma GLM + two-part models	All 50 states + District of Columbia

Patients with anxiety spend \$2,600 to \$3,000 more per year across all three cardiovascular conditions.

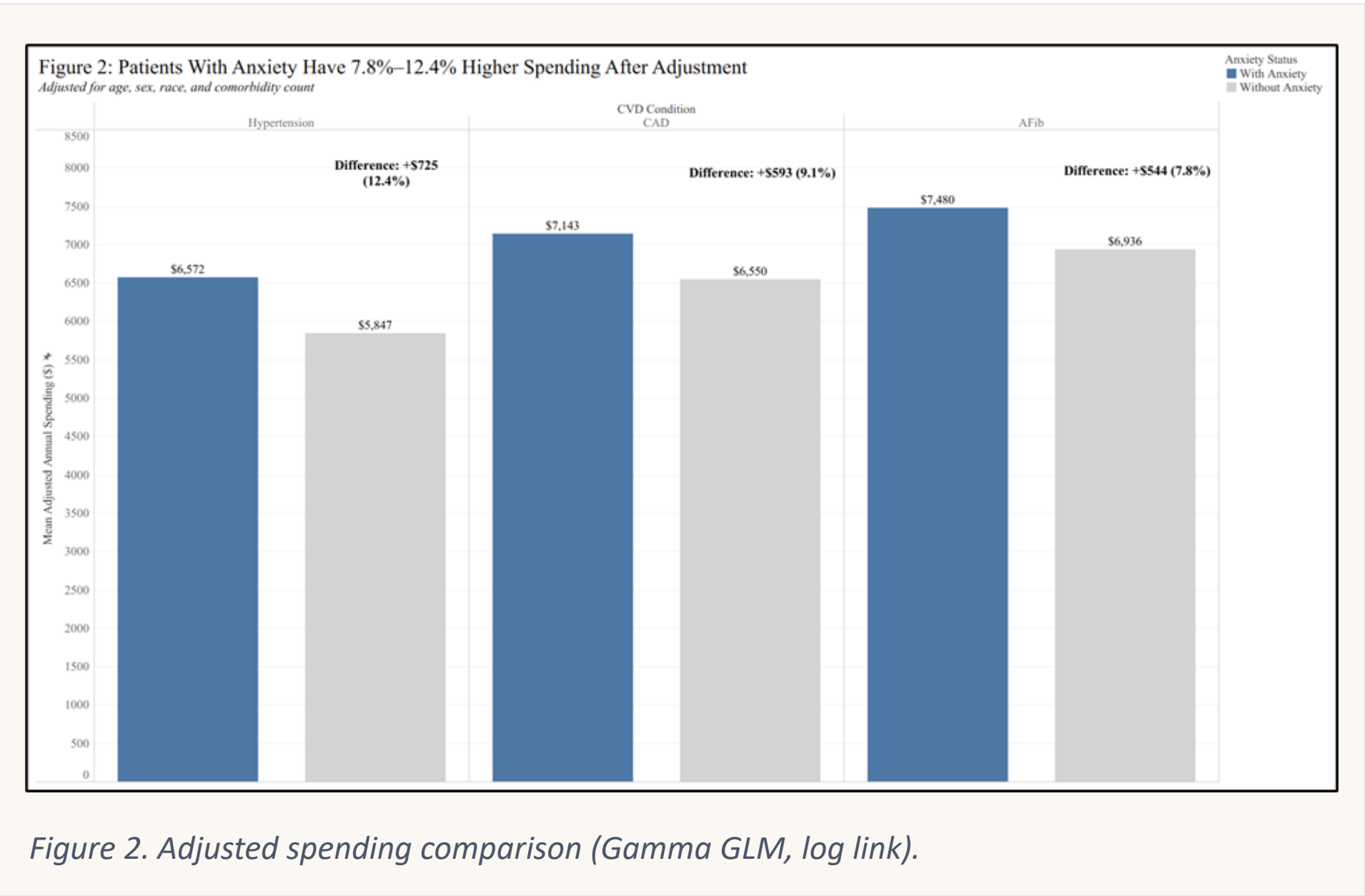


RAW SPENDING GAP, BY CONDITION		
CONDITION	DIFFERENCE	HIGHER SPENDING
Hypertension	+\$2,967	+63.1%
CAD	+\$2,621	+47.4%
AFib	+\$2,620	+44.6%

CAVEAT

These are raw differences. The gap could reflect sicker patients rather than an independent anxiety effect, motivating the adjusted analysis.

Anxiety independently increases spending by 7.8% to 12.4% after controlling for age, sex, race, and comorbidities.



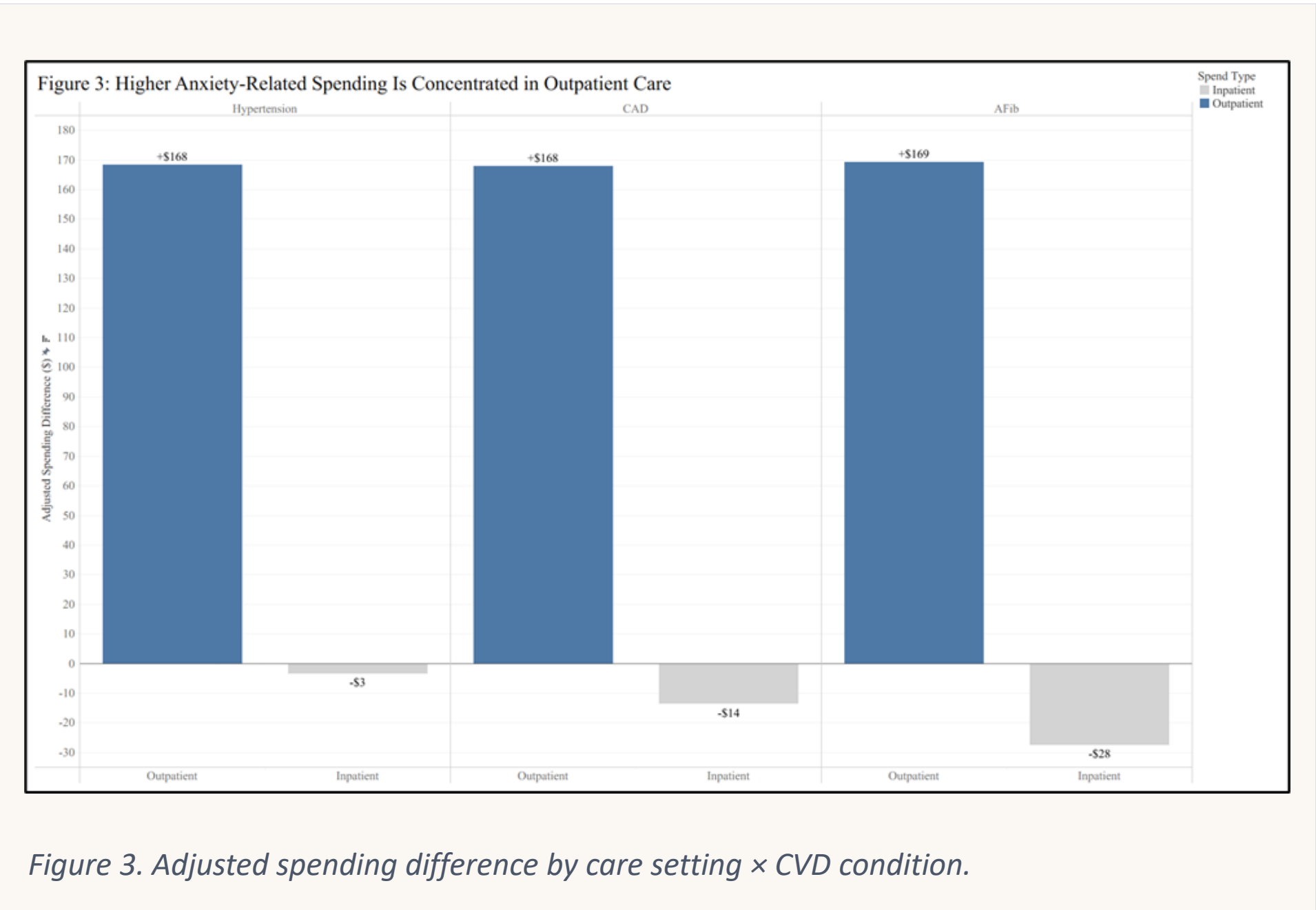
Source: CMS DE-SynPUF · Gamma GLM, log link · adjusted for age, sex, race, comorbidity count

ADJUSTED COST RATIOS (95% CI)		
CONDITION	COST RATIO	HIGHER SPENDING
Hypertension	1.124	+12.4%
CAD	1.090	+9.0%
AFib	1.078	+7.8%

INTERPRETATION

Adjusted spending gap of **about 7.8% to 12.4%**. The gap narrows but does not disappear. Anxiety has an independent association with spending beyond patient characteristics. All $p < 0.001$.

Higher anxiety-related spending is concentrated in outpatient care, not inpatient care.



HOW THE GAP BREAKS DOWN

OUTPATIENT

+\$168 to +\$169

Higher adjusted outpatient spending across hypertension, CAD, and AFib among patients with anxiety.

INPATIENT

Minimal

Adjusted inpatient spending differences are small and close to zero across all three conditions.

SUPPORT FROM TWO-PART MODELS

Patients with anxiety had **43% to 58% higher odds of any outpatient use**, but no significant difference in hospitalization odds.

All 50 States and DC show a positive anxiety spending gap. A national pattern, not a regional one.

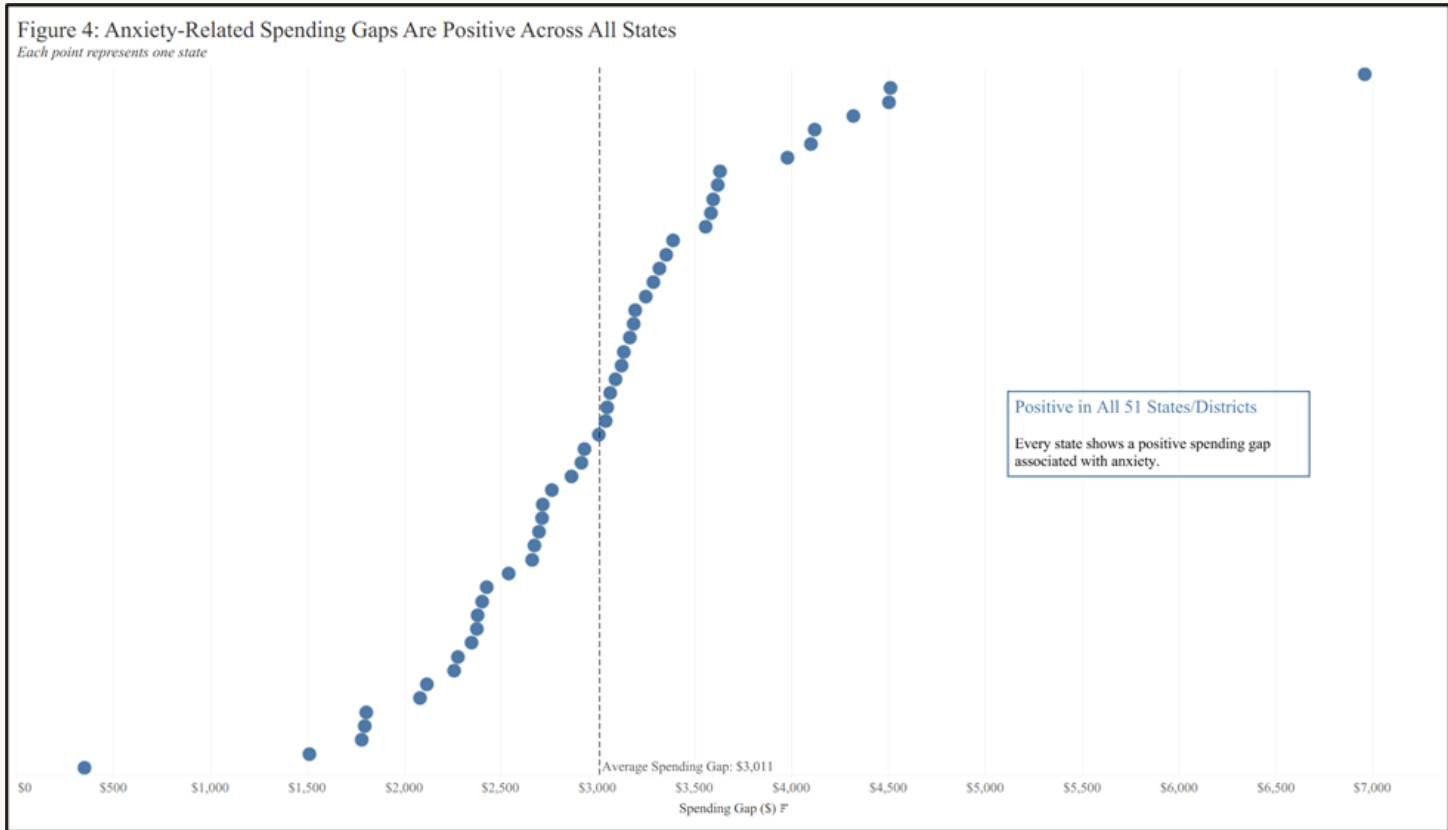


Figure 4. Distribution of state-level spending gaps. Each point = one state.

STATE-LEVEL DISTRIBUTION

STATES WITH POSITIVE GAP
51 / 51

MEAN STATE-LEVEL GAP
\$3,011

RANGE
\$353 to \$6,959

COEFFICIENT OF VARIATION
0.32

ONE-SAMPLE T-TEST
t = 22.24 p < 0.001

LARGEST GAP BY CONDITION
Hypertension

POLICY TAKEAWAY

Integrating behavioral health screening into cardiovascular care management may help reduce excess outpatient healthcare use and better support value-based care in Medicare.